Skills Summary

Factoring Cubes and Higher-Degree Expressions

Use this reference while completing your worksheet or when studying.

Skill 1 — Sum and difference of cubes

Patterns: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

Signs (SOAP): the binomial keeps the **S**ame sign; the trinomial's middle term takes the **O**pposite sign; its last term is **A**lways **P**ositive.

Cubes on sight: 1, 8, 27, 64, 125, 216. The trinomial never factors again over the integers.

Example: Factor $x^3 + 27$
Step 1. Two terms, both perfect cubes, so this is a cube pattern — not something to group or treat as a quadratic.
Step 2. Here $a = x$ and $b = 3$. The sum pattern gives the binomial $x + 3$, then $x^2 - 3x + 9$.
$\Rightarrow$ $\boldsymbol{(x + 3)(x^2 - 3x + 9)}$

Try it: Factor $x^3 - 64$

check: $\boldsymbol{(x - 4)(x^2 + 4x + 16)}$

Skill 2 — Factor by grouping

When: four terms. Split them into two pairs, pull the GCF out of each pair, and the two leftover binomials must *match* — that match is what you factor out. If they differ only in sign, take -1 out of the second pair.

Then keep going: the remaining quadratic is often a difference of squares.

Example: Factor $x^3 - 3x^2 - 4x + 12$
Step 1. Four terms with no factor common to all of them, so I try grouping in pairs.
Step 2. $x^2(x - 3) - 4(x - 3) = (x - 3)(x^2 - 4)$, and $x^2 - 4$ still splits.
$\Rightarrow$ $\boldsymbol{(x - 3)(x - 2)(x + 2)}$

Try it: Factor $x^3 + 5x^2 - x - 5$

check: $\boldsymbol{(x + 5)(x - 1)(x + 1)}$

Skill 3 — Quadratic in disguise

When: the exponents come in the pattern $2n$ and n, as in $x^4 - 10x^2 + 9$ (here $n = 2$). Substitute $u = x^n$, factor the ordinary quadratic in u, then put x^n back.

You are not finished there: each bracket in x^2 is usually a difference of squares and has to be split too.

Example: Factor $x^4 - 10x^2 + 9$
Step 1. The exponents are four and two, so $u = x^2$ turns this into a quadratic I already know how to factor.
Step 2. $u^2 - 10u + 9 = (u-1)(u-9)$, so back in x it is $(x^2-1)(x^2-9)$ — and both brackets split.

$\Rightarrow$ $\boldsymbol{(x-1)(x+1)(x-3)(x+3)}$

Try it: Factor $x^4 - 29x^2 + 100$

check: $\boldsymbol{(x-2)(x+2)(x-5)(x+5)}$

Skill 4 — From factors to zeros and shape

Rule: once a polynomial is a product of factors, set each factor equal to zero — the zero-product property turns one hard equation into several easy ones.

A factor of x contributes the zero $x = 0$. A repeated factor makes the graph *touch* the axis; a single factor makes it *cross*, so the sign flips there.

Example: Real zeros of $x^3 - x^2 - 6x$
Step 1. I want the zeros, and a factored form hands them over, so I factor completely first.
Step 2. $x(x^2 - x - 6) = x(x-3)(x+2)$, so each of the three factors gives one zero.

$\Rightarrow$ $\boldsymbol{x = -2,\ 0,\ 3}$

Try it: Real zeros of $x^3 - 9x$

check: $\boldsymbol{x = -3,\ 0,\ 3}$